PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Electrical Engineering Exam: Introduction to Circuits
Instructions: Please answer all questions to the best of your ability. Show your work where applicable.

Section 1: Multiple Choice (2 points each, 8 points total)

Instructions: Choose the BEST answer for each question.

Question 1: What principle is this lab primarily designed to reintroduce?
A) Ohm's Law
B) Kirchhoff's Laws
C) Superposition Principle
D) Thevenin's Theorem

Question 2: According to the text, what components are connected in series on the left side of the circuit?
A) Resistors and a diode
B) Resistors and an inductor
C) Resistors and a capacitor
D) Inductors and a capacitor

Question 3: In Figure 2b, what voltage values are supplied to the resistors on the right side of the circuit?
A) 5 volts and 0 volts
B) 5 volts and -5 volts
C) 10 volts and -10 volts
D) 0 volts and -5 volts

Question 4: According to the experimental results, which table corresponds to the circuit with a shorted capacitor?
A) Table E1
B) Table E2
C) Table E3
D) Table P1

Section 2: Short Answer (4 points each, 12 points total)

Instructions: Answer each question in 2-3 sentences.

Question 5: Briefly describe the objective of this lab in your own words.

Question 6: Explain what the superposition principle is, according to the text.

Question 7: What discrepancies does the report mention when comparing the simulated MultiSim waveform to Graph E3?

Section 3: Problem-Solving (10 points each, 30 points total)

Instructions: Provide detailed solutions and explanations for each problem.

Question 8: Based on the provided information, if the resistance of the 560 ohm resistor was actually 570 ohms, how would this affect the VI (rms) value in Table E1? Explain your reasoning.

Question 9: Using the values provided in Tables E1 and E2, and assuming that the superposition principle holds perfectly, calculate the expected VO (RMS) value. Show all steps in your calculation. How does this calculated value compare to the VO (RMS) in Table E3, and what does this indicate?

Question 10: The report mentions a percent error of 11.37% when comparing Table P1 and Table E3 input RMS voltage. Without knowing the values of Table P1 calculate what the input RMS value in Table P1 would be based on this percent error and Table E3's input RMS voltage.